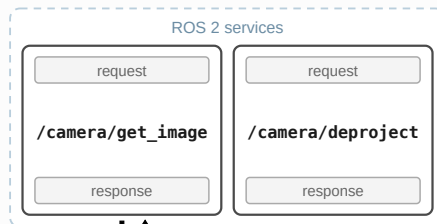


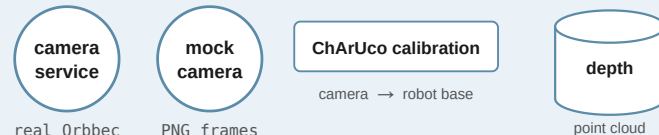
SENSE

reading the diagram

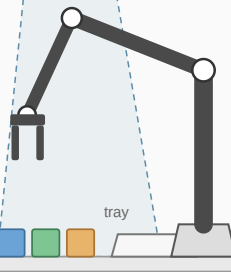
- ROS 2 node
- ROS 2 service (request / response)
- deterministic code-grounding (§V)
- configuration, logging, supervision



vlm_camera_service



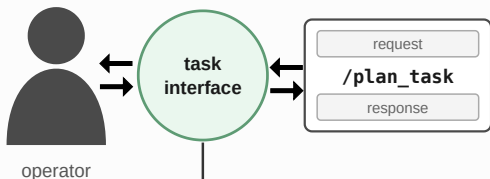
physical rig



UR5cb · OnRobot RG2

PLAN

task_interface_node
CLI · web GUI



robo_reason_planner



code-grounding · plan time → Cases 1-4
_fix_object_height · _fix_release_height (LLM)
depth deprojection · _apply_grasp_width (VLM)

robo_reason_reasoning

EmbodiedAgent

7 reasoning methods
fhp · ffhp · react · cot_sc · tot · always_act · self_refine

LLM client

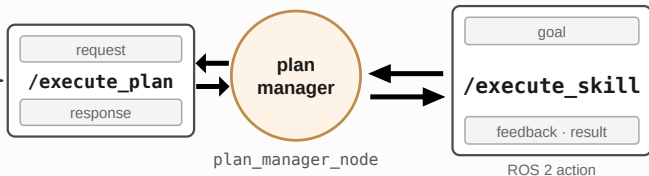
text + scene JSON

VLM client

image-grounded

EmbodiedAgentsPrompts · backend configurable (kimi-k2.6, §VI)

ACT



plan_manager_node
grounding · placement time
distribute_zone_releases
→ Case 6 (static half only)

robo_reason_executor



real robot



dry-run

skills

approach · pick · release
move_home · wait

code-grounding · execution time

tcp_offset_z_for_width (width-dependent flange-to-contact offset) → Case 5

config.py

one pydantic-settings object: model, reasoning method,
grounding mode, timeouts, robot IP — all env-overridable

debug_recorder

per-run artifact: command, active config, raw model response,
camera frame, execution log → the basis of the §IV diagnosis

robo_reason_gui

supervises the stack, the UR driver and the camera as child
processes · live retuning · emergency stop (CancelExecution)